

1. Project Overview

This project analyzes Human Resource data by combining employee, attendance, performance and payroll information. The data was cleaned using Microsoft Excel and visualized in Power BI to create an interactive People Analytics Dashboard. The dashboard helps HR teams track workforce trends, employee performance, attendance, and compensation, supporting better and faster decision-making.

Objective

To clean and analyze HR data using Microsoft Excel and Power BI, and develop an interactive People Analytics Dashboard that provides insights into employee demographics, attendance, performance, and payroll, enabling data-driven HR decision-making and effective workforce management. .

2. Data Sources

- Source Description and Timeline: Google Dataset Search and 2023-2025.
- Domain: Human Resource.

3. Problem Statement

HR data such as employee details, attendance, performance, recruitment, and payroll records are often stored in multiple spreadsheets, making analysis and reporting time-consuming. This project aims to consolidate these datasets into a single interactive Power BI dashboard to provide insights into workforce demographics, attendance trends, employee performance, payroll costs, and attrition, helping HR teams make informed and data-driven decisions.

4. Attribute (Column /Features) Details:

Employees Table

Attribute Name	Data Type	Description
Employee_ID	Text	Unique Employee Identifier
Candidate_ID	Text	Candidate Reference Number
Gender	Text	Employee Gender
Marital_Status	Text	Marital Status
Employment_Type	Text	Permanent, Intern, Contract
Work_Mode	Text	WFH, Hybrid, WFO
Department	Text	Employee Department
Designation	Text	Job Role
Location	Text	Office Location
Age	Numeric	Employee Age
DOJ	Date	Date of Joining
Employment_Status	Text	Active or Terminated
CTC	Currency	Annual Compensation
Engagement_Score	Decimal	Employee Engagement Score
Satisfaction_Score	Decimal	Employee Satisfaction Score

Experience	Text	Total Experience
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Attendance Table

Attribute Name	Data Type	Description
Employee_ID	Text	Unique Employee Identifier
Date	Date	Attendance Date
Month & Year	Text	Reporting Month and Year
Present_Days	Integer	Number of Days Present
Absent_Days	Integer	Number of Days Absent
LOP_Days	Integer	Leave Without Pay Days
Overtime_Hours	Decimal	Total Overtime Hours Worked
Late_Mark_Count	Integer	Number of Late Attendance Instances

Performance Table

Attribute Name	Data Type	Description
Employee_ID	Text	Unique Employee Identifier
Review_Year	Integer	Performance Review Year
Manager_Rating	Decimal	Performance Rating Given by Manager
KPI_Score	Decimal	Key Performance Indicator Score
Training_Hours	Integer	Total Training Hours Completed
Productivity_Score	Decimal	Employee Productivity Assessment Score
Promotion_Count	Integer	Number of Promotions Received

Payroll Table

Attribute Name	Data Type	Description
Employee_ID	Text	Unique Employee Identifier
Date	Date	Payroll Processing Date
Month & Year	Text	Payroll Month and Year
Gross_Pay	Currency	Total Gross Salary Paid
Bonus	Currency	Incentive or Bonus Amount
PF	Currency	Provident Fund Contribution
ESI	Currency	Employee State Insurance Contribution
Net_Pay	Currency	Final Salary Paid After Deductions

5. Tools & Technologies

- **Excel:** Data Cleaning, Missing Value Handling, Formatting, Derived Columns Creation, Initial Data Validation, Pivot Table Analysis.
- **Power BI:** Data Modelling, Relationship Creation ,DAX Calculations, Dashboard Development, KPI Monitoring, Interactive Reporting

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

Data Cleaning: The dataset was cleaned by removing duplicate records, handling missing values, standardizing date formats, and correcting inconsistent entries to improve data quality and accuracy.

Data Transformation: Additional analytical fields such as Engagement Band, Satisfaction Band, Experience Band, and Stability Group were created to enable better workforce analysis and reporting.

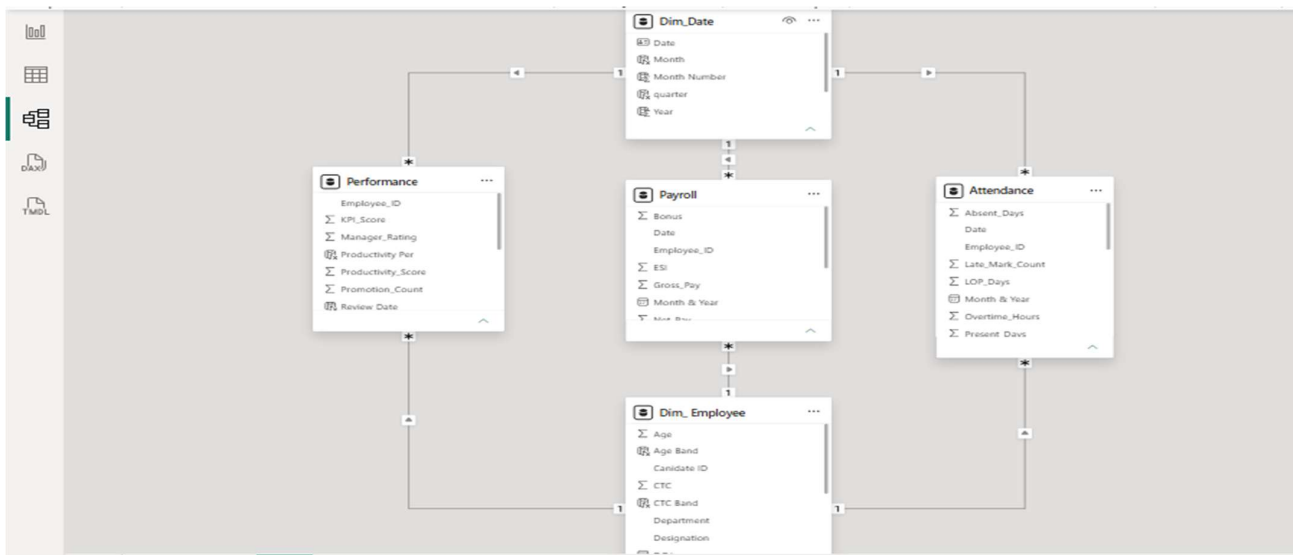
Filtering and Sorting: The data was organized and structured through filtering and sorting techniques to simplify analysis and enhance reporting efficiency.

Data Segmentation: The dataset was divided into separate subject-based tables, including Employee, Attendance, Recruitment, Performance, and Payroll, to support effective data modeling and analysis.

7. Data Modelling and DAX (Power BI)

- **Data Model:** Relationships were established in Power BI between the Employees, Attendance, Payroll, Performance, and Date Dimension tables to create a structured data model. This enabled seamless analysis, cross-filtering, and accurate reporting across different HR functions.

Note : Fact Table (Performance, Payroll, Attendance) & Dimension Table (Employee)



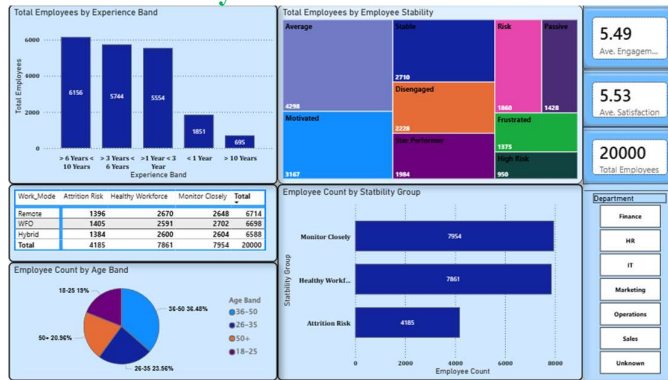
Calculated Columns & DAX Measures: Several DAX measures were created to calculate key HR metrics, including employee count, attrition rate, attendance statistics, payroll costs, engagement and satisfaction scores, performance indicators, and promotion trends. These measures provide meaningful insights to support workforce analysis and decision-making.

8. Analysis and Visualizations (Power BI)

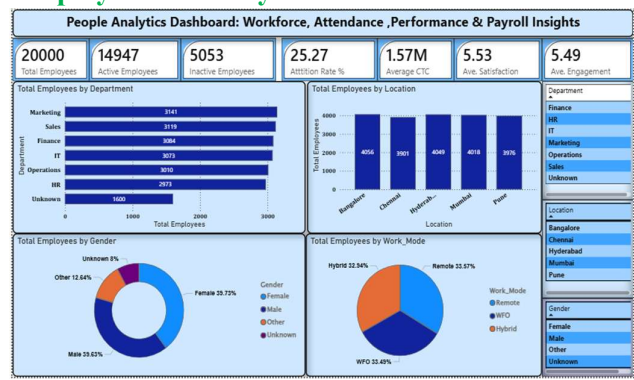
Dashboard Features:

- Developed multiple dashboard pages, including Employee Summary, Workforce Analysis, Attendance Analysis, Payroll Analysis, and Performance Analysis, to provide comprehensive HR insights.
- Designed KPI cards to track metrics such as Total Employees, Active Employees, Attrition Rate, Average CTC, Engagement Score, Attendance Percentage, Payroll Cost, Productivity Score, and Promotions.
- Created visualizations including Bar Charts, Line Charts, Donut/Pie Charts, Tables, and KPI Cards to analyze workforce demographics, attendance trends, compensation patterns, and employee performance.
- Enhanced report usability through Slicers, Dynamic KPI Cards, Clear Visual Titles, Data Labels, and Interactive Dashboard Navigation for seamless data exploration

Workforce Analysis



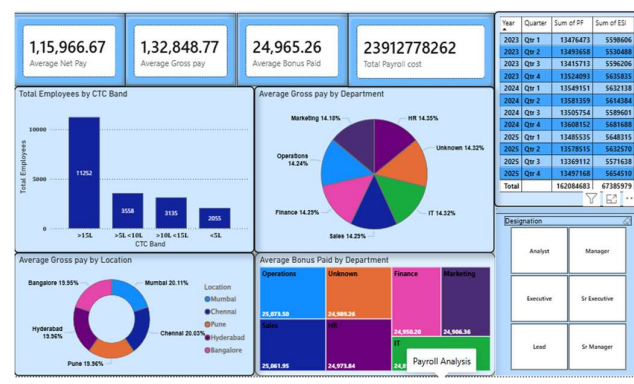
Employee Summary



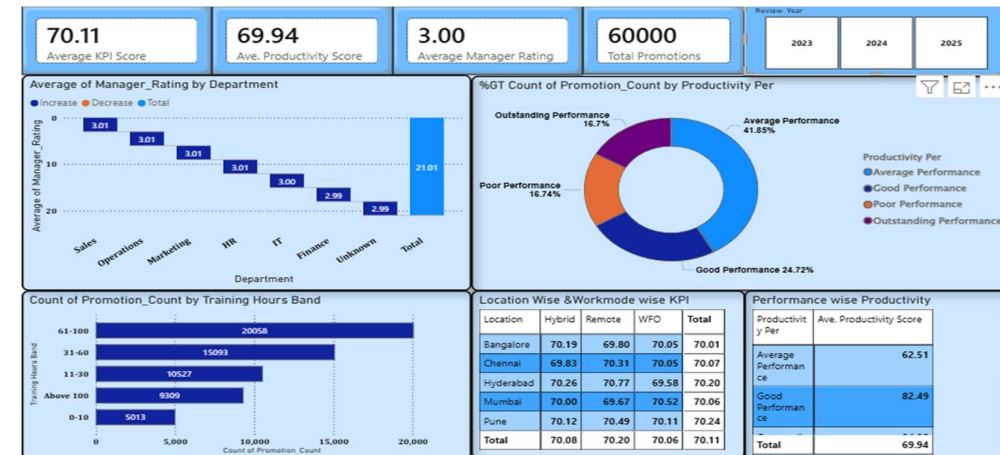
Attendance Analysis



Payroll Analysis



Performance Analysis



9. Insights & Conclusions

- Key Findings:** Summarize trends, patterns, or anomalies identified in the data

Descriptive Insights

- Workforce composition varies significantly across departments and locations.
- Employee satisfaction scores are generally moderate to high.

- Attendance trends indicate opportunities for reducing absenteeism.

Diagnostic Insights

- Higher overtime hours are associated with specific departments.
- Attrition patterns can be linked with engagement and stability indicators.

Predictive Insights

Employees with low engagement and satisfaction scores may be at higher risk of turnover.

Prescriptive Insights

- HR teams should focus on employee engagement initiatives.
- Attendance improvement programs can reduce absenteeism.
- Performance-based learning interventions may enhance productivity

10. Conclusions

The integration of Microsoft Excel and Power BI provided an efficient end-to-end analytics solution for workforce management. The People Analytics Dashboard successfully transformed raw HR datasets into meaningful insights, enabling organizations to monitor employee performance, optimize payroll management, improve attendance tracking, and support strategic human resource planning through data-driven decision-making.